BIOS 731 final report

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```r
library(here)
```

```
## here() starts at C:/Users/xdong63/OneDrive - Emory/Desktop/Fall 2024 Emory/BIOS 731/BIOS 731 final p
```

```r
library(readr)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.3
```

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

Overview

This document summarizes the results of the CGM missing-interval prediction analysis. We report cross-validated performance for onset prediction and duration prediction, and we summarize full-data Bayesian predictions using the previously generated result tables and figures saved in the project directories.

##Data sources used in this summary

The following previously created files are used in this report:

`results/metrics/real_data_cv_results.csv` `results/metrics/real_data_cv_summary.csv` `outputs/tables/cv_summa`
`outputs/tables/top20_bayesian_onset_risk.csv` `outputs/tables/bayes_onset_deciles.csv`
`outputs/figures/cv_bar_metrics.png` `outputs/figures/cv_fold_metrics.png` `outputs/figures/cv_rmse.png`
`outputs/figures/bayes_onset_prob_hist.png` `outputs/figures/bayes_onset_decile_plot.png`
`outputs/figures/bayes_duration_scatter.png` `Cross-validated results` `Fold-level results`

Cross-validated results

Fold-level results

```
cv_results <- read_csv(here::here("results", "metrics", "real_data_cv_results.csv"), show_col_types = F
cv_results
```

```
## # A tibble: 10 x 7
##    method      auroc    auprc rmse_duration coverage_95  fold error_message
##    <chr>       <dbl>    <dbl>         <dbl> <lgl>       <dbl> <lgl>
##  1 glm_em      0.647  0.00516         925.  NA              1 NA
##  2 bayes_mcmc  0.647  0.00522          NA   NA              1 NA
##  3 glm_em      0.910  0.178           238.  NA              2 NA
##  4 bayes_mcmc  0.905  0.178            NA   NA              2 NA
##  5 glm_em      0.518  0.0569          237.  NA              3 NA
##  6 bayes_mcmc  0.515  0.0646           NA   NA              3 NA
##  7 glm_em     NA     NA                NA   NA              4 NA
##  8 bayes_mcmc NA     NA                NA   NA              4 NA
##  9 glm_em      0.655  0.00192          17.2 NA              5 NA
## 10 bayes_mcmc  0.660  0.00218          NA   NA              5 NA
```

Table 1 reports fold-specific performance for the fitted models. The most relevant metrics for onset prediction are AUROC and AUPRC. RMSE is shown for duration prediction when available.

```
kable(cv_results, digits = 3, caption = "Table 1. Fold-specific cross-validated results.")
```

Table 1: Table 1. Fold-specific cross-validated results.

method	auroc	auprc	rmse_duration	coverage_95	fold	error_message
glm_em	0.647	0.005	924.813	NA	1	NA
bayes_mcmc	0.647	0.005	NA	NA	1	NA
glm_em	0.910	0.178	238.405	NA	2	NA
bayes_mcmc	0.905	0.178	NA	NA	2	NA
glm_em	0.518	0.057	236.845	NA	3	NA
bayes_mcmc	0.515	0.065	NA	NA	3	NA
glm_em	NA	NA	NA	NA	4	NA
bayes_mcmc	NA	NA	NA	NA	4	NA
glm_em	0.655	0.002	17.245	NA	5	NA
bayes_mcmc	0.660	0.002	NA	NA	5	NA

The cross-validated results suggest that onset prediction is feasible, although performance varies substantially across folds. The AUROC values indicate that the model has some ability to rank windows preceding a missing interval higher than windows not followed by missingness. The AUPRC values are lower, which is expected in a rare-event setting. Duration prediction is less stable, with RMSE varying across folds and some folds lacking usable duration estimates.

Visual summaries of cross-validation

Mean performance by method

```
knitr::include_graphics(here::here("outputs", "figures", "cv_bar_metrics.png"))
```

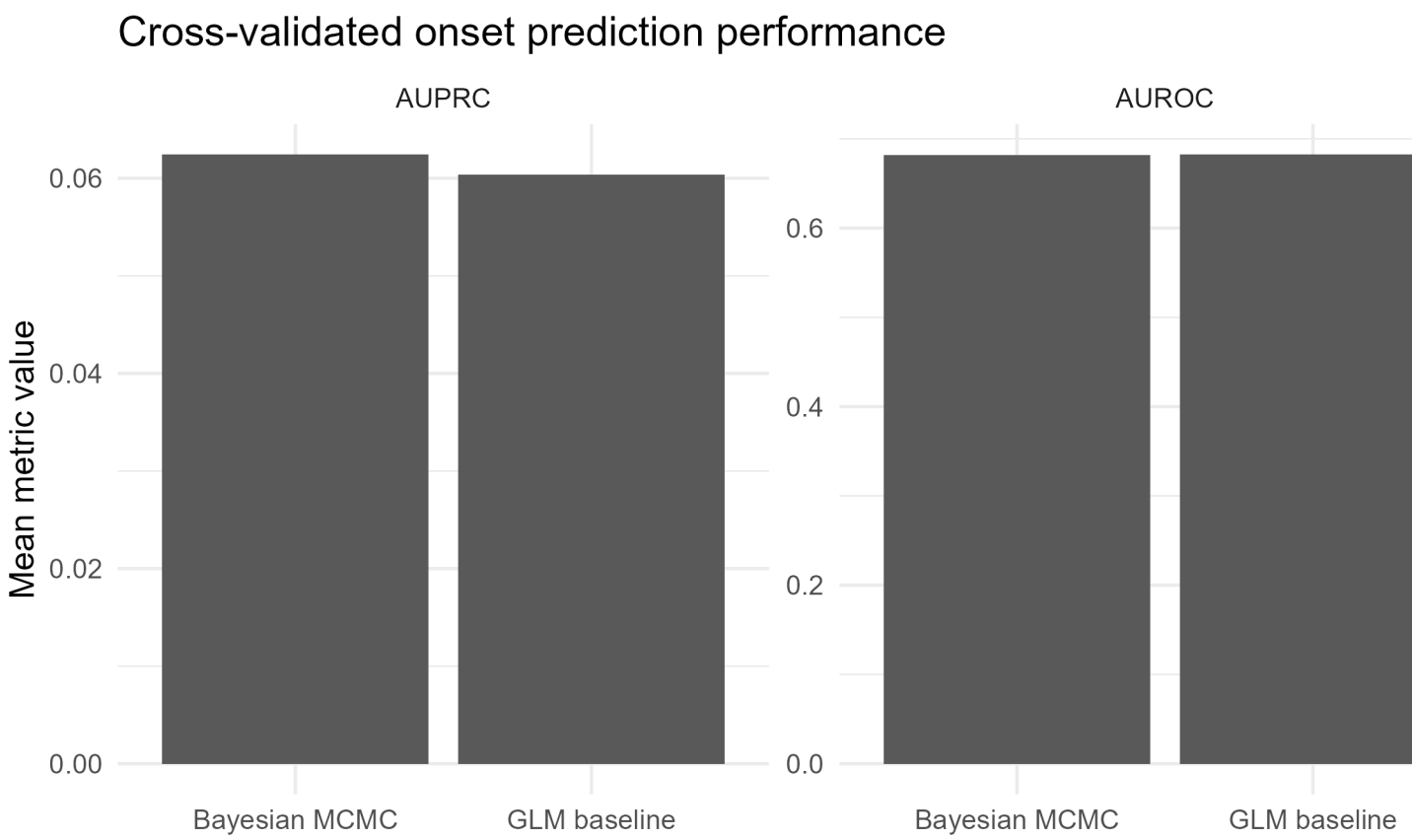


Figure 1 summarizes mean cross-validated onset prediction performance across methods. AUROC and AUPRC are shown separately to highlight both ranking ability and rare-event precision.

Fold-specific performance

```
knitr::include_graphics(here::here("outputs", "figures", "cv_fold_metrics.png"))
```

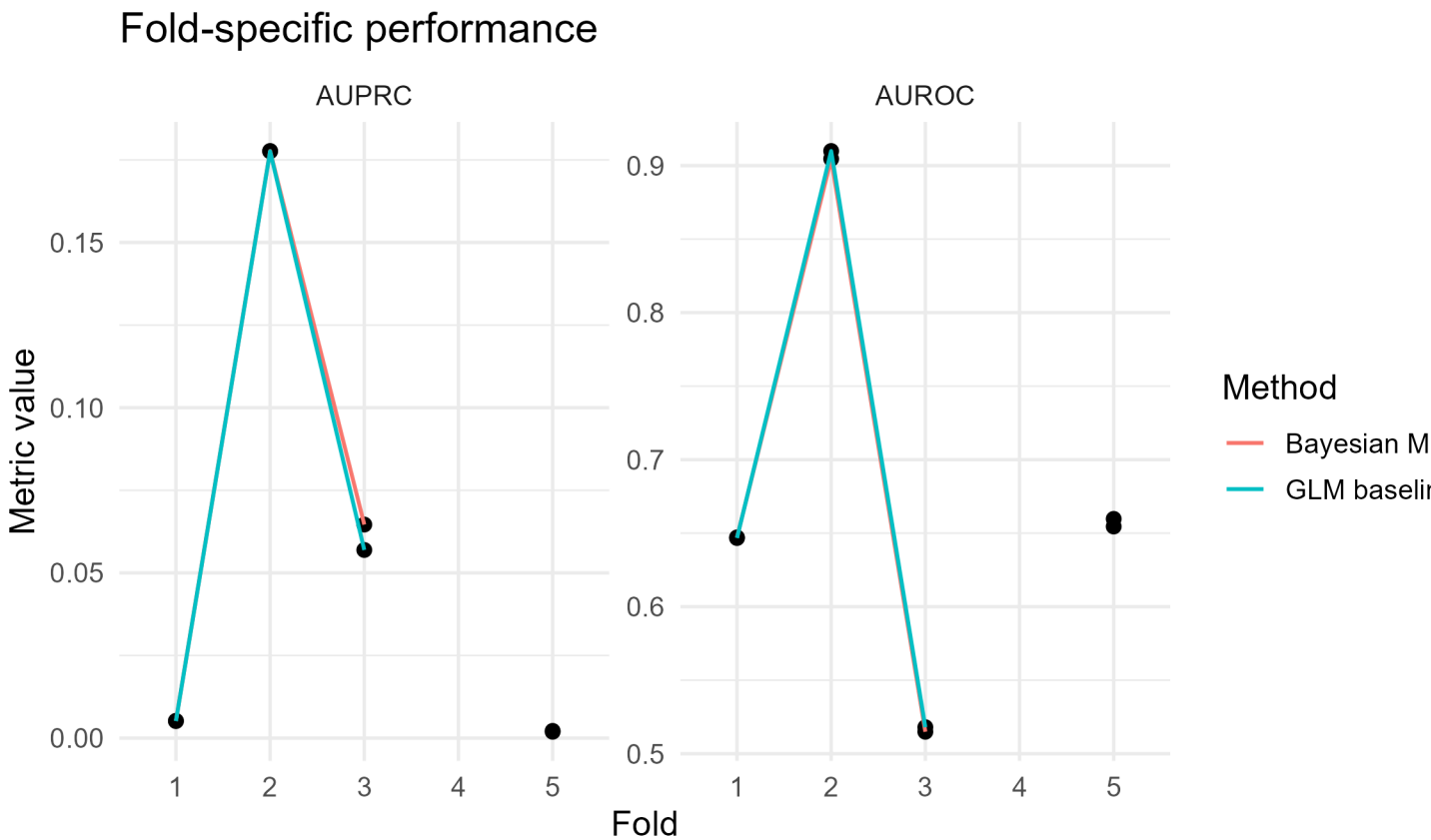
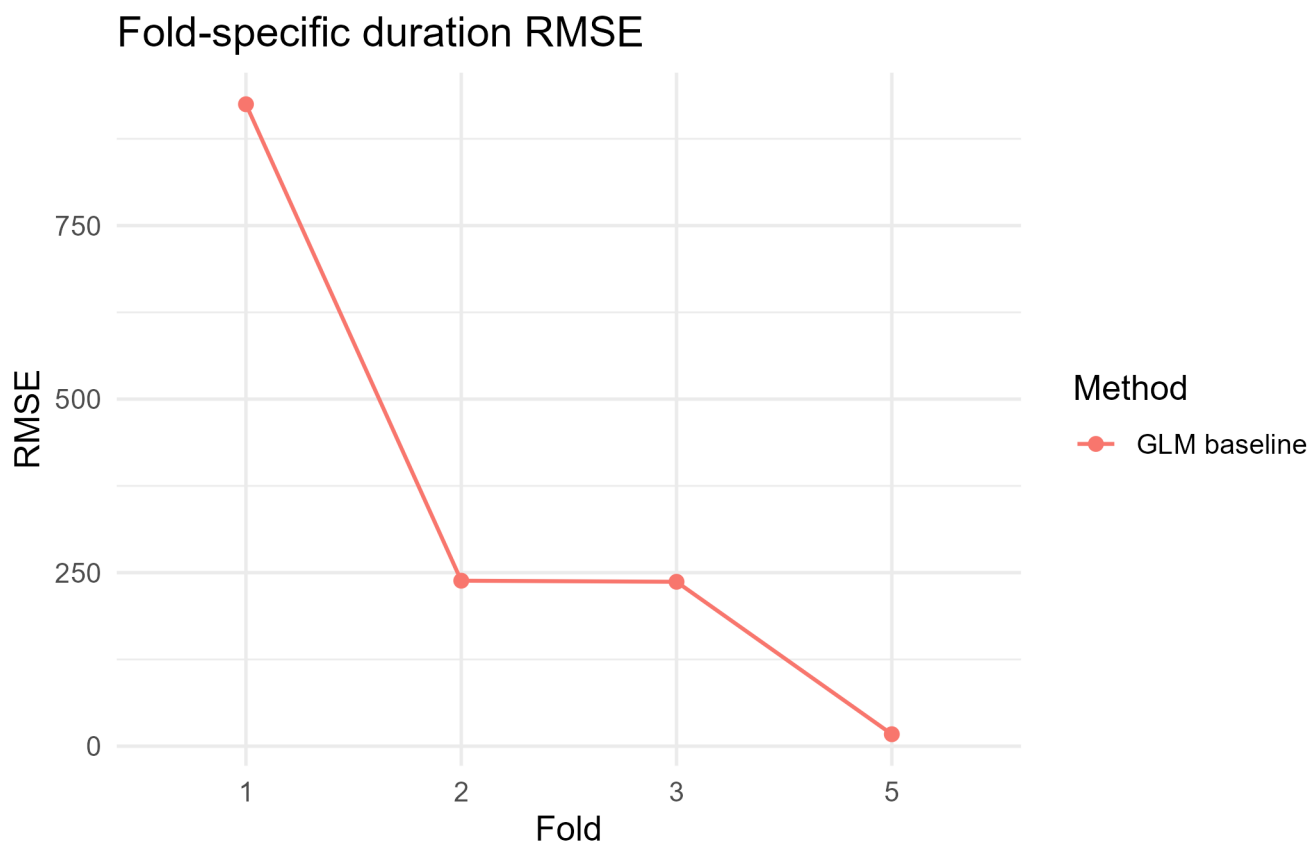


Figure 2 shows that performance is heterogeneous across folds. This variation is consistent with the rarity and uneven distribution of missing-interval onset events across patients and folds.

Duration RMSE

```
knitr::include_graphics(here::here("outputs", "figures", "cv_rmse.png"))
```



When duration RMSE is available, the results suggest that duration estimation is more difficult and less stable than onset prediction.

Full-data Bayesian onset predictions

Distribution of predicted onset probabilities

```
knitr::include_graphics(here::here("outputs", "figures", "bayes_onset_prob_hist.png"))
```

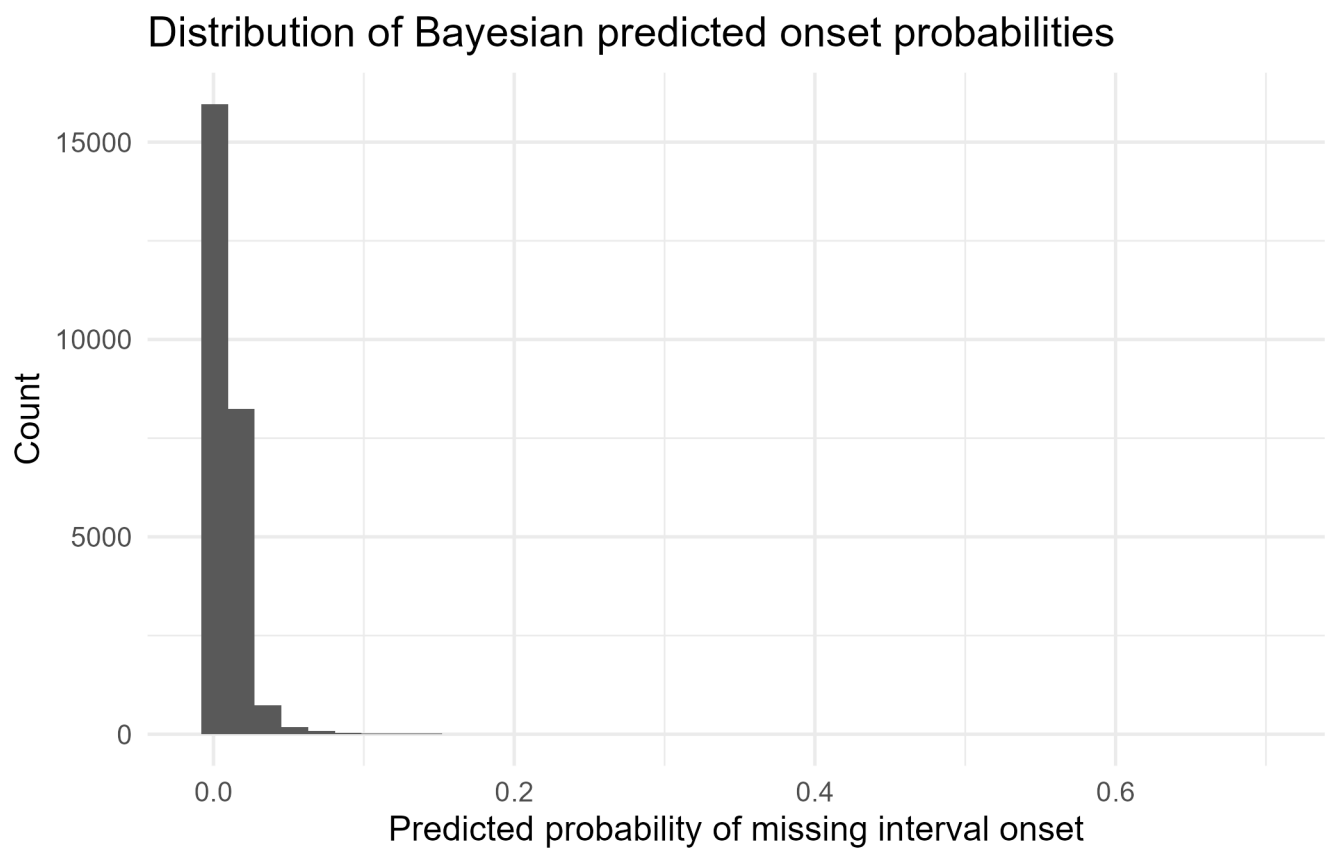


Figure 4 displays the distribution of predicted probabilities from the full-data Bayesian onset model. Most windows are expected to have low predicted probability, since missing-interval onset is a relatively uncommon event.

Observed onset rate by risk decile

```
if (file.exists(here::here("outputs", "tables", "bayes_onset_deciles.csv"))) {
  onset_deciles <- read_csv(here::here("outputs", "tables", "bayes_onset_deciles.csv"), show_col_types =
  kable(onset_deciles, digits = 4, caption = "Table 3. Observed onset rate by predicted-risk decile.")
}
```

Table 2: Table 3. Observed onset rate by predicted-risk decile.

risk_decile	n	observed_onset_rate	mean_predicted_prob
1	2534	0.0000	0.0008
2	2534	0.0008	0.0010
3	2534	0.0000	0.0010
4	2534	0.0000	0.0012
5	2534	0.0012	0.0015
6	2534	0.0079	0.0041
7	2534	0.0028	0.0100
8	2533	0.0134	0.0123
9	2533	0.0170	0.0161

risk_decile	n	observed_onset_rate	mean_predicted_prob
10	2533	0.0422	0.0377

```
knitr::include_graphics(here::here("outputs", "figures", "bayes_onset_decile_plot.png"))
```

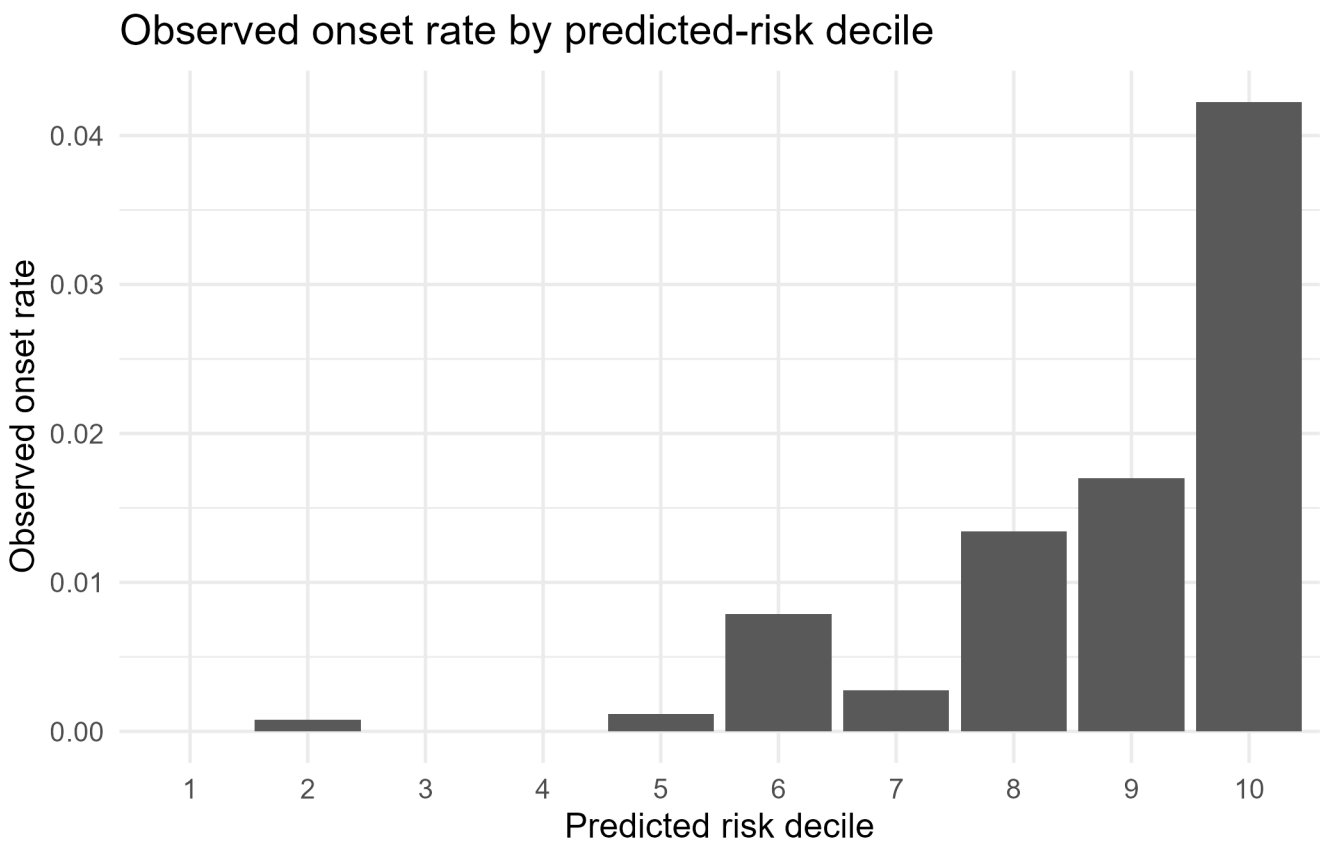


Table 3 and Figure 5 assess practical risk stratification. If the model is useful, higher predicted-risk deciles should correspond to higher observed onset rates.

Highest-risk windows

```
if (file.exists(here::here("outputs", "tables", "top20_bayesian_onset_risk.csv"))) {
  top_risk <- read_csv(here::here("outputs", "tables", "top20_bayesian_onset_risk.csv"), show_col_types
  kable(top_risk, digits = 4, caption = "Table 4. Top 20 windows ranked by Bayesian predicted onset risk
}
```

Table 3: Table 4. Top 20 windows ranked by Bayesian predicted onset risk.

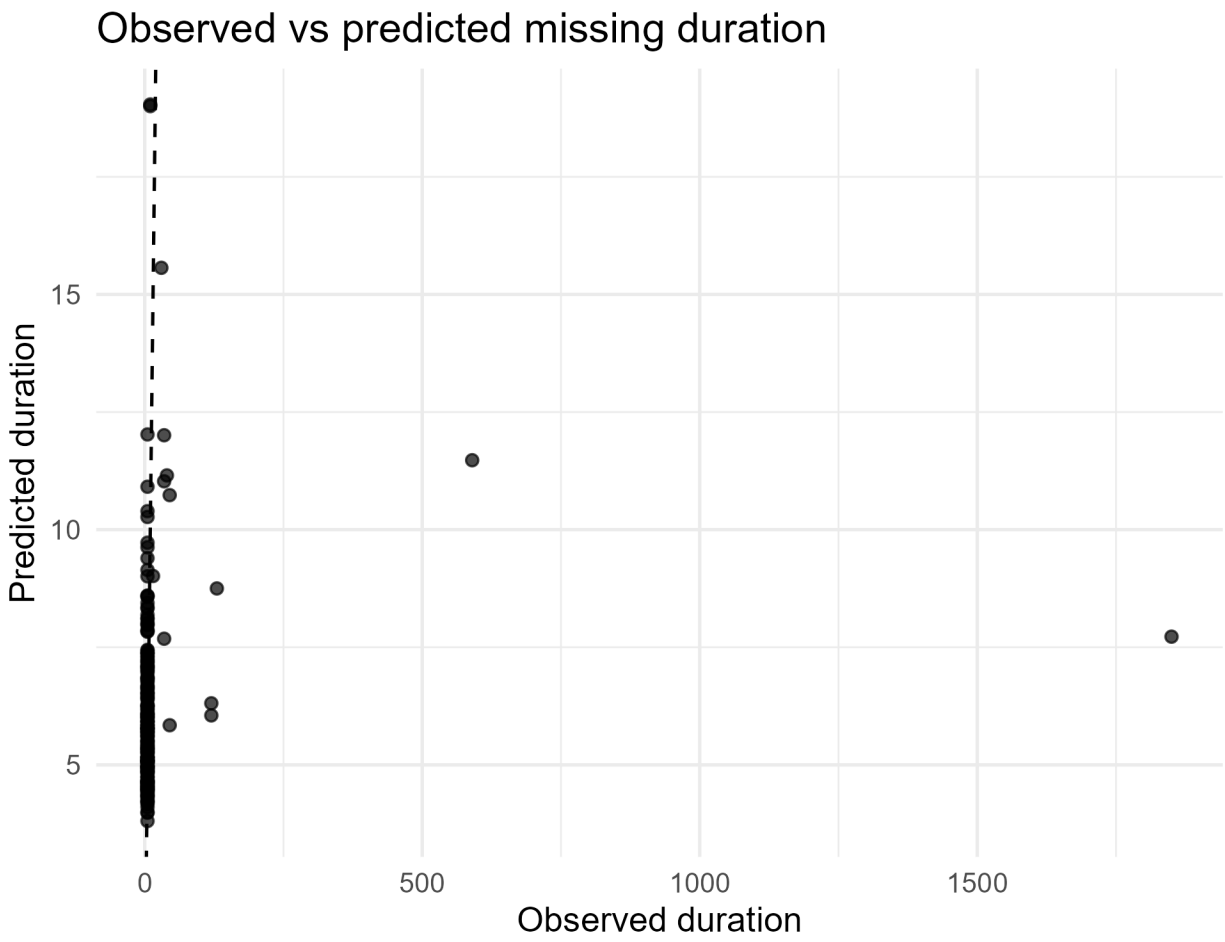
patient_id	end_time	y_onset	onset_prob
EM036	170	0	0.6946
EM036	190	0	0.6620

patient_id	end_time	y_onset	onset_prob
EM036	160	1	0.6554
EM036	180	1	0.5932
UOM050	2875	0	0.5771
EM036	175	0	0.5675
UOM050	2870	0	0.5550
UOM027	2965	0	0.4752
UOM027	2960	0	0.4516
UOM050	2880	0	0.3952
G021_extra	1135	0	0.3852
EM036	460	1	0.3786
UOM051	565	0	0.3354
G072	1260	1	0.3205
UOM051	570	0	0.3161
UOM051	560	0	0.3034
UOM050	2865	0	0.3029
G021_extra	195	0	0.3024
EM024_extra	4165	0	0.2955
G021_extra	1145	0	0.2950

Table 4 lists the windows assigned the highest predicted probability of imminent missingness by the full-data Bayesian onset model.

Full-data Bayesian duration predictions

```
knitr::include_graphics(here::here("outputs", "figures", "bayes_duration_scatter.png"))
```



When available, Figure 6 compares observed and predicted missing durations. Agreement around the 45-degree line indicates better predictive performance. In practice, duration prediction is often noisier than onset prediction, so this figure should be interpreted with caution.

Interpretation

Overall, the results indicate that predicting the onset of a missing interval is more successful than predicting its duration. Cross-validated AUROC values suggest meaningful discriminatory ability, while AUPRC values reflect the added difficulty of a rare-event setting. The Bayesian full-data onset model provides an additional way to rank windows by future missingness risk, and the decile-based summaries help assess whether higher-risk windows indeed correspond to higher observed onset rates. By contrast, duration prediction remains more unstable, both across folds and in full-data modeling.